

Deployment Options for KakaoBank

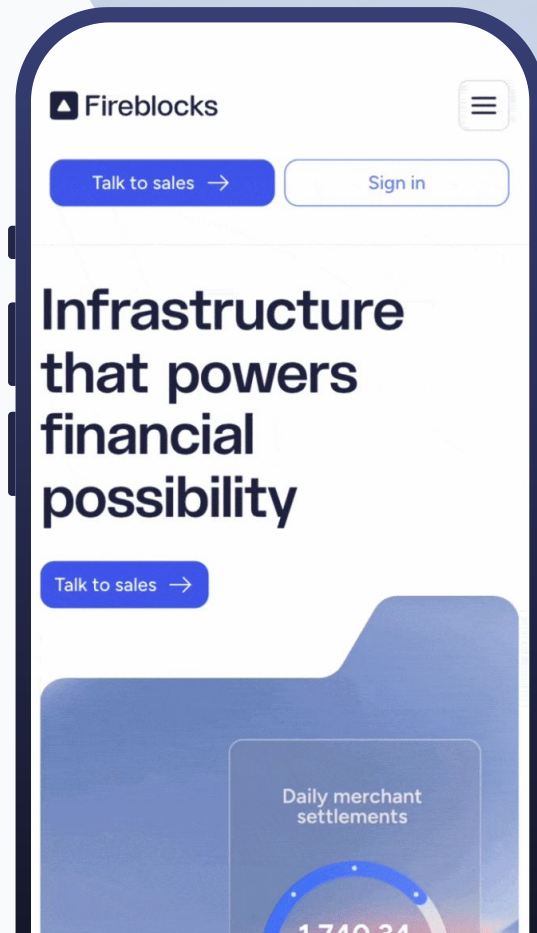


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KeyLink PaaS HSM Deeper Dive

Detailed HSM overview as an alternative deployment option.



Why we proposed this architecture

Key Requirements

What any proposed option has to satisfy

- ▲ This service will be defined as material workloads including key management and signing
- ▲ A custody platform Kakao can **develop**, **deploy**, **operate** and **manage** in its own application.
- ▲ It is a key requirement the bank builds the application layer on top of Fireblocks in a PaaS model.

Under Korean regulation, a workload's permissible scope is set by its service classification, and that classification follows architectural facts, not a self-assigned label. So we tested each option against the statutory definition of a platform.

KakaoBank's second requirement names the same four functions the statute does.

The Cloud Computing Act's PaaS definition

Develop | **Deploy** | **Operate** | **Manage**

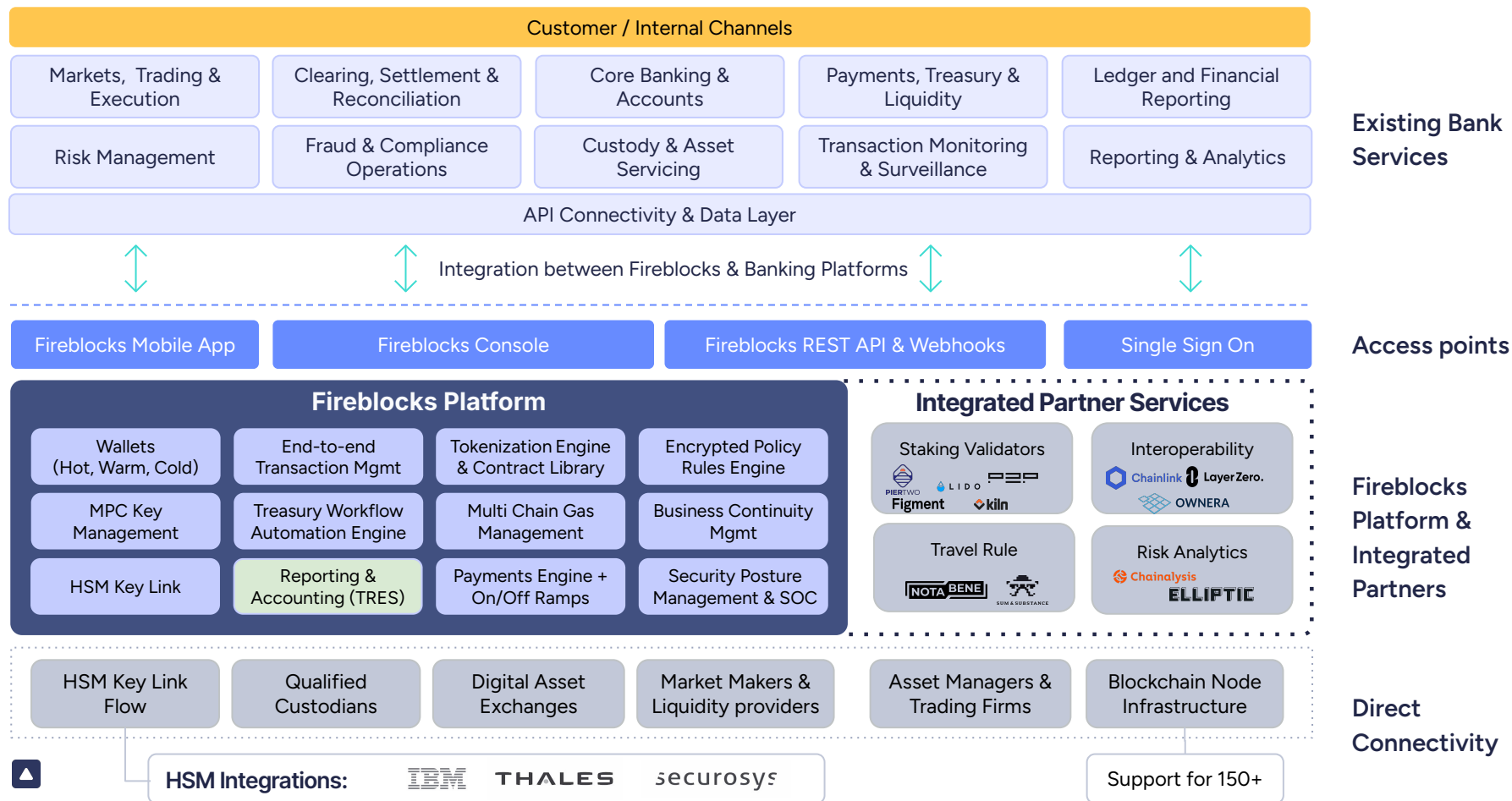
Every option that follows is read against these four.

Deployment options for key management

Options	Option 1: Cloud PaaS MPC	Option 2: KeyLink PasS HSM	Option 3: Korean Local Instance
Delivery Timelines	Days to weeks	2-3 Months	4+ Months
Criteria	Used in large scale deployments Removes the single point of compromise for the private key, by generating and storing the key using 3 unique key shares Two shares held with Fireblocks in US Data Center. Controlling Key share held with KakaoBank in local Korean VPC. Backup kit, which includes all 3 key shares encrypted can be stored locally in Korea.	HSM Device stored locally in Kakao environment. Requires integration development & thorough testing to connect Fireblocks to HSM More complex deployment model when compared to Cloud PaaS MPC	Consider this section as optional roadmap items. Korean Local Instance - If required the Cloud PaaS MPC model could be co-located in South Korea. This would mean the two Fireblocks key shares would also be in region as well as core services.



How Fireblocks tech stack fits into your organization in detail



Flexible Architecture Options

The next four slides set out the Fireblocks PaaS architecture for Kakao Bank.

01

Fireblocks PaaS model

The high-level architecture, and where the Kakao Bank application layer meets the Fireblocks platform.

02

Deeper dive

The same architecture with its network connections, protocols and ports.

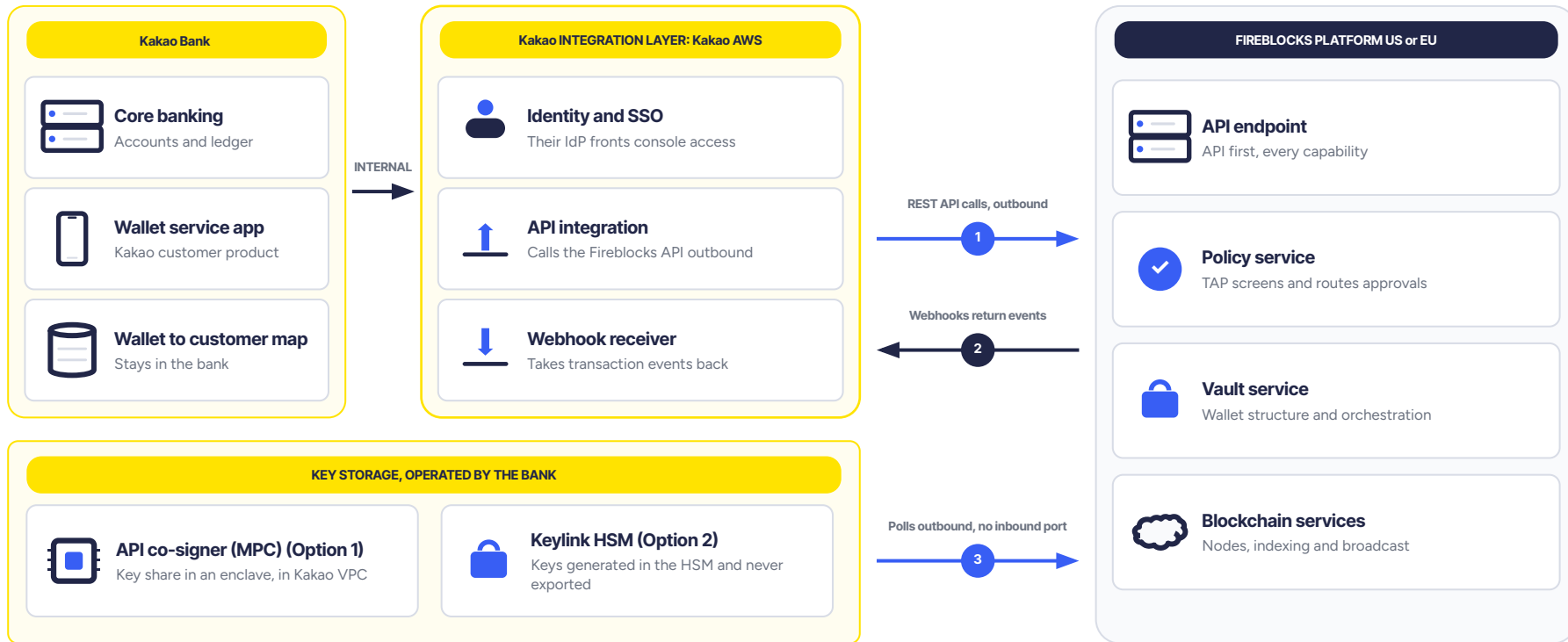
03

Network requirements

The endpoints, ports and directions the KakaoBank network team provisions.

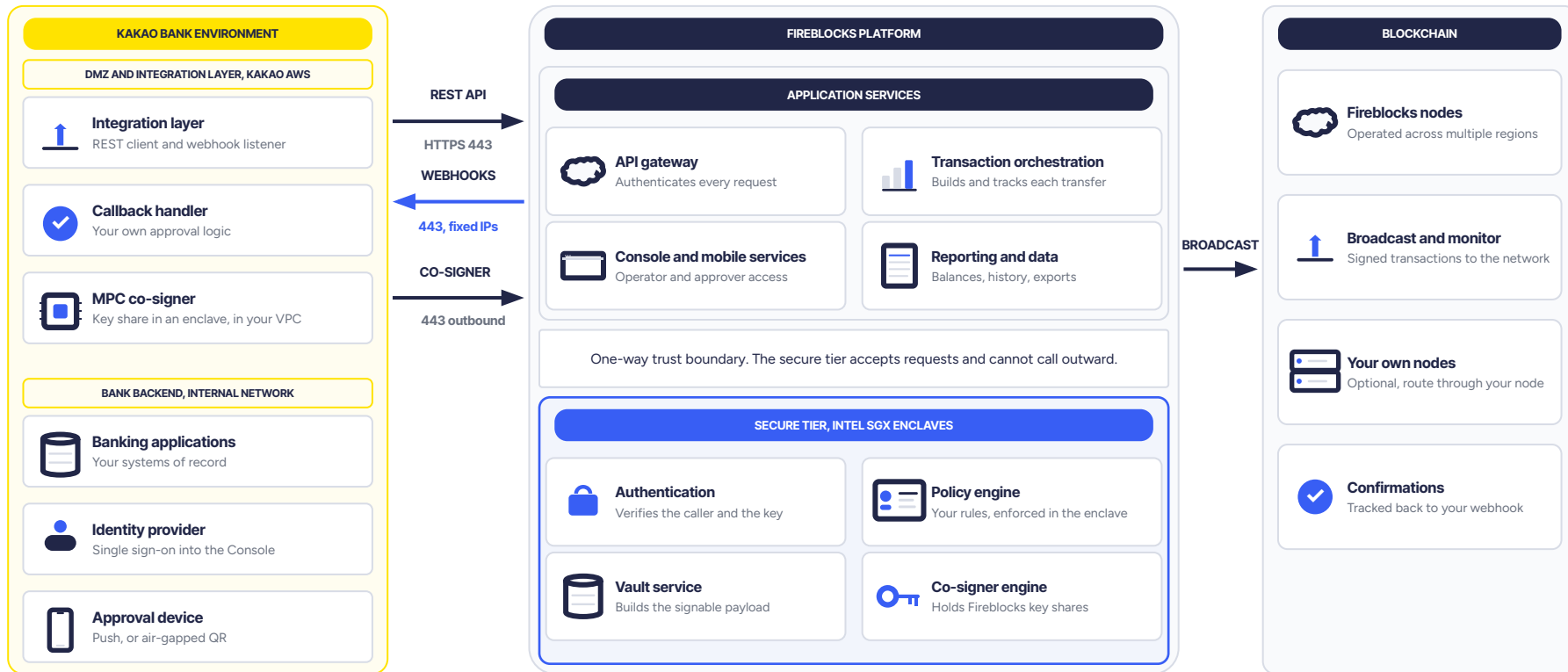


Option 1: Cloud PaaS MPC - High Level View



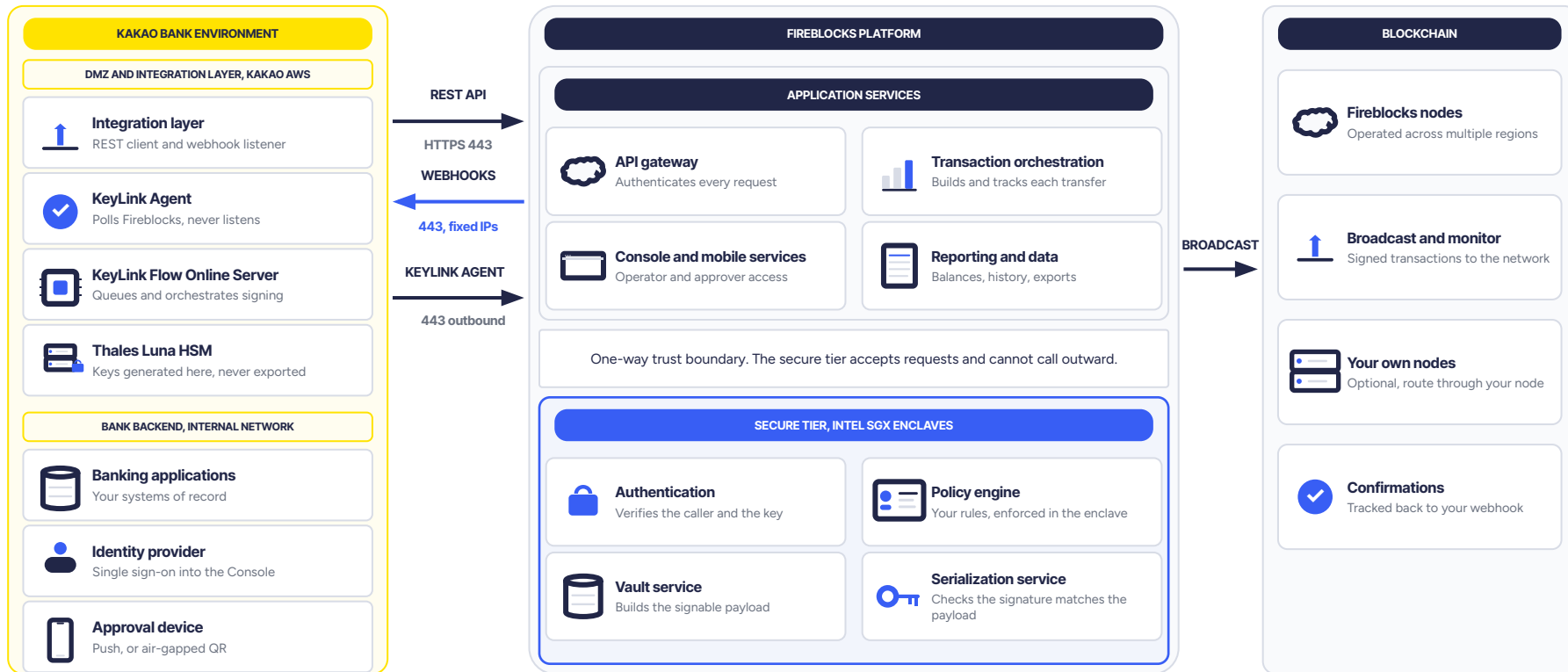
Option 1: Cloud PaaS MPC: Deeper Dive & Network Connections

Every connection to Fireblocks runs over TLS. The webhook listener is the only inbound path into your network.



Option 2: Key Link PaaS HSM

Keys are generated in your HSM and never leave it. Fireblocks holds only the public key.



What your network team needs to open

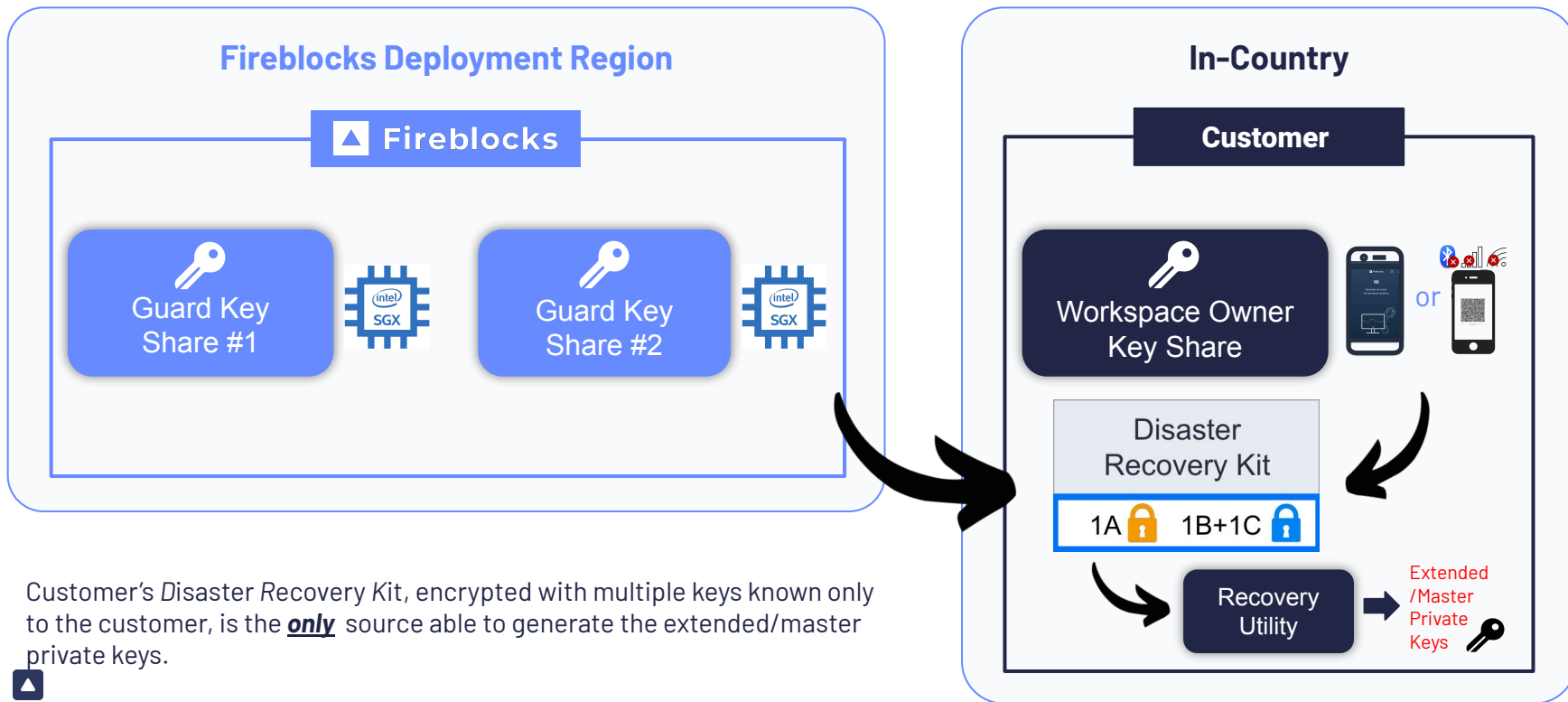
Ports, direction and endpoints for a Fireblocks PaaS deployment.

Flow	Direction	Protocol and port	Endpoint	Notes
REST API	Outbound	HTTPS 443	api.fireblocks.io, or the EU endpoints	API key header plus a request token signed with RS256
Console	Outbound	HTTPS 443	console.fireblocks.io	Operator browsers, single sign-on to your identity provider
Webhooks	Inbound	HTTPS 443	3.135.57.98, 63.179.12.78, 3.77.138.100	Signed with RS512. Verify against the published key endpoint
Co-signer, running	Outbound	HTTPS and WSS 443	mobile-api, signurl and s3signurl	Holds a WebSocket open, falls back to long polling. No inbound port
Co-signer, updates	Outbound	443, and 5000 to the registry	Fireblocks storage and package mirrors	Maintenance windows only
Callback handler	Within your network	HTTPS, port of your choosing	Co-signer to your approval service	Answer inside 30 seconds or the request fails
Approval device	Outbound	HTTPS 443, and push	Fireblocks and the push provider	Cold wallet approvals stay air-gapped and travel by QR
Time sync	Outbound	NTP 123	Your integration hosts	Request tokens live 30 seconds, so clocks have to be in step



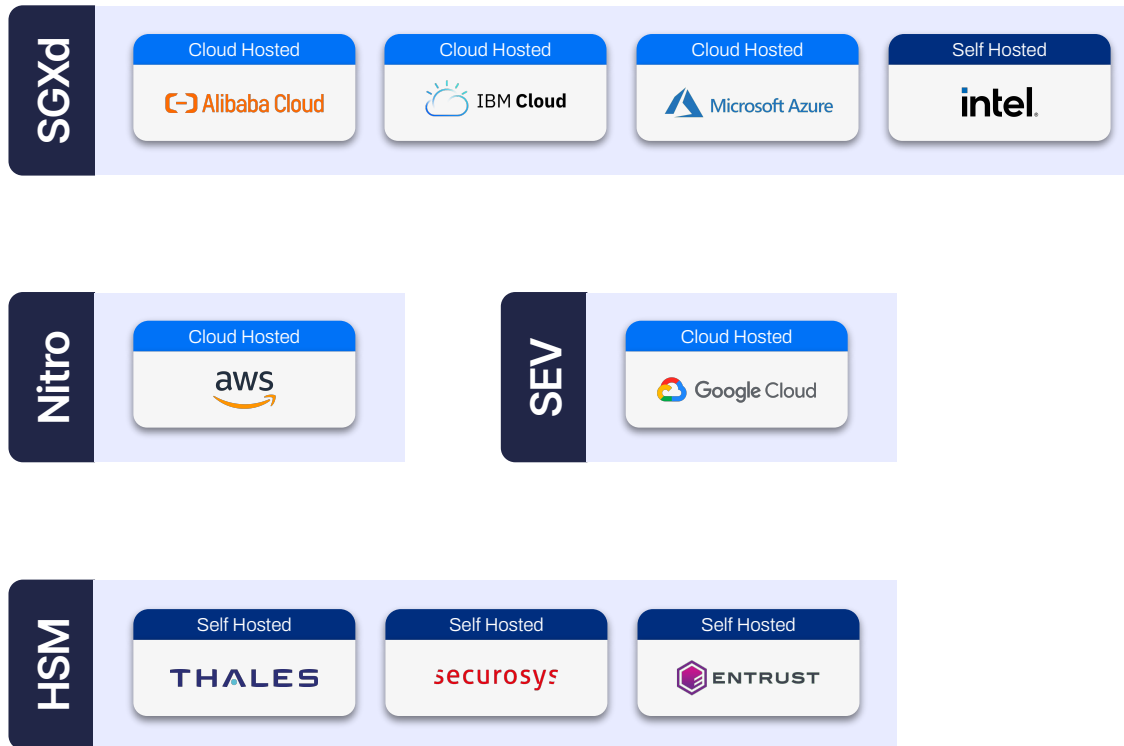
Your Keys, Your Assets, In Country

Customer's only complete MPC-CMP key set (i.e. Disaster Recovery Kit) is stored in-country/on-shore.s



Hardware Isolation Options

Choose the Trusted Execution Environment technology option that works best for the organization. These can be deployed in your local cloud instance.



HSM Deployment



What is Fireblocks Keylink Flow?

A secure, scalable, and compliant digital asset infrastructure designed for modern financial institutions.

01

An end-to-end HSM powered Digital Asset Infrastructure

02

Allows banks and financial institutions to integrate their existing KMS

03

Provides a rich governance and approval flow layered with FIPS Hardware Security

04

Addresses the slow and non-compliant nature of existing cold storage

05

Built for scalability, powering the most demanding financial applications



Fireblocks & Thales Partnership

What this partnership brings to our customers

- **Full Key Ownership & Regulatory-Grade Control**

Custodians can maintain complete, provable ownership of private keys inside their own Thales HSMs while still orchestrating digital-asset workflows through Fireblocks — giving them a control model that aligns with supervisory expectations in highly regulated jurisdictions.

- **Best-in-Class HSM Security**

The integration unites hardware-anchored security (Thales) with approvals-based operational protection (Fireblocks), delivering a hybrid architecture that exceeds traditional custody models and provides layered defense against insider threats, key-extraction attacks, and operational compromise.

- **Seamless Integration Into Existing Custody Infrastructure**

Custodians can extend their existing HSM-based key management stacks into digital-asset operations without re-architecting or migrating away from hardware systems they already trust, reducing deployment friction and accelerating time-to-market for new asset services.

- **Scalable, Institutional-Grade Transaction Orchestration**

The partnership provides custodians with mature APIs, policy-driven workflows, and automation tools that enable high-throughput settlement, complex governance controls, and multi-entity signing workflows tailored for institutional operations.

- **Access to the Broader Fireblocks Network & Ecosystem**

Custodians benefit from Fireblocks' extensive network of exchanges, liquidity venues, payment rails, and counterparties — allowing them to offer clients faster transfers, more asset support, and integrated financial-services connectivity while keeping all signing within Thales hardware.

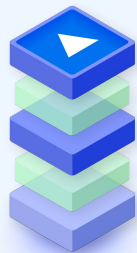


Fireblocks KeyLink Flow - Online HSM Signing

Hot Wallet HSM signing for Payment ready solutions

Fireblocks

SaaS Operational Stack



- Transaction Engine
- Transaction Approval Policy Engine (TAP)
- H/A Blockchain Infrastructure
- Blockchain Indexes
- Serialization Engine



Customer Production Environment



KeyLink Agent

Fireblocks
KeyLink
Flow
Online
Server

LUNA Client



THALES
Building a future we can all trust



Blockchain Networks

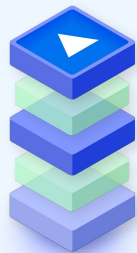


Fireblocks KeyLink Flow - Offline HSM Signing

Cold Wallet HSM signing for Payment ready solutions

Fireblocks

SaaS Operational Stack



- Transaction Engine
- Transaction Approval Policy Engine (TAP)
- H/A Blockchain Infrastructure
- Blockchain Indexes
- Serialization Engine



Blockchain Networks



Production Environment



KeyLink
Agent

**Fireblocks
KeyLink
Flow**
Online Server



Air-gapped Environment

**Fireblocks
KeyLink
Flow**
Offline Server

LUNA
Client



THALES
Building a future we can all trust